

WOOSANG KIM

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RESEARCH INTERESTS

Human-Computer Interaction, Perception, Multisensory Interaction, Haptics, Food Texture Perception, Interactive Interfaces, Intraoral Interfaces, Electrotactile Stimulation, Sensing, Virtual and Mixed Reality

EDUCATION

The University of Electro-Communications **Tokyo, Japan**

M.Eng. in Informatics

Department of Informatics, Media Science and Engineering Program Apr. 2024 – Mar. 2026

Thesis: “Intraoral Electrical Stimulation for Evoking Oral Sensations and Modulating Food Texture Perception”

Advisor: Takuya Nojima

The University of Electro-Communications **Tokyo, Japan**

B.Eng. in Informatics and Computer Engineering

Department of Informatics, Media Science and Engineering Program Apr. 2020 – Mar. 2024

Thesis: “Study on a Method for Stimulation of Periodontal Ligament Receptors by Transcutaneous Electrical Stimulation”

Advisor: Takuya Nojima

RESEARCH EXPERIENCE

Nojima Laboratory, The University of Electro-Communications **Tokyo, Japan**

Graduate / Undergraduate Researcher Apr. 2023 – Mar. 2026

- Investigated intraoral electrical stimulation and its effects on sensation and food texture perception.
- Developed real-time experimental systems for gingival electrotactile stimulation using custom hardware.
- Designed and conducted human-subject studies on intraoral electrical stimulation, including experiments on chewing-synchronized stimulation.

Advisor: Takuya Nojima

HCI Lab, University of Chicago **Chicago, IL, USA**

Summer Research Intern Jul. 2024 – Aug. 2024

- Developed and evaluated electrical stimulation patterns to characterize sensations evoked by intranasal trigeminal stimulation.

Advisors: Pedro Lopes, Jas Brooks

PUBLICATIONS & PRESENTATIONS

Under Review

Woosang Kim, Sho Sakurai, Koichi Hirota, and Takuya Nojima. “An Exploratory Study of Chewing-Synchronized Intraoral Electrotactile Stimulation for Modulating Texture-Related Perception During Mastication.” IEEE Transactions on Haptics, 2026.

Conference Presentations

Woosang Kim, Sho Sakurai, Koichi Hirota, and Takuya Nojima. (2026). [*Characterizing Gingival Electrical Stimulation Toward Intraoral Texture Display*](#), Poster presented at IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), Daegu, Republic of Korea.

Woosang Kim, Shiori Iwahama, Sho Sakurai, Koichi Hirota, and Takuya Nojima. (2023). *Investigation of Oral Sensory Information for Selective Stimulation of Periodontal Ligament Receptors*, The Virtual Reality Society of Japan, Tokyo, Japan.

Woosang Kim, Sho Sakurai, Koichi Hirota, and Takuya Nojima. (2023). *Investigation of Transcutaneous Electrical Stimulation to Periodontal Ligament Receptors and the Oral Sensory Information Generated*, The Virtual Reality Society of Japan SIG Haptics, Miyagi, Japan.

WORK EXPERIENCE

NTT DOCOMO, INC.

Tokyo, Japan

Spatial Data Scientist

Apr. 2026 – Present

- Analyze geospatial population data and support infrastructure operations for Mobile Spatial Statistics, DOCOMO's population statistics platform.

PROJECTS & LEADERSHIP

UEC Idea Demonstration Contest

Tokyo, Japan

Co-leader

May. 2023 – Dec. 2023

- Developed a mask-based interface that detects mouth movements; the project received an award from Aflac.
- Coordinated hardware development, programming, and project management as co-leader.

UT-Virtual (A VR Club based at the University of Tokyo)

Tokyo, Japan

Member

Sep. 2020 – Mar. 2022

- Developed VR projects and showcased them at CIC Tokyo.
- Organized an international exchange with the University of Maryland XR Club.

SCHOLARSHIP

Summer Research internship Funding (\$4,500)

Aug. 2024

Hirose Scholarship Foundation (\$15,000/ year)

Apr. 2023 – Mar. 2024

Hasegawa International Scholarship Foundation (\$8,200/ year)

Apr. 2021 – Mar. 2023

TEACHING EXPERIENCE

The University of Electro-Communications

Tokyo, Japan

International Student Tutor

Jul. 2025 – Sep. 2025

- Supported Martin Mathieu Heudron during his research internship in the Nojima Lab, providing guidance on laboratory tools and assisting with research activities.

The University of Electro-Communications

Tokyo, Japan

Teaching Assistant

Jun. 2025 – Aug. 2025

Advanced Human Interface, Instructor: Takuya Nojima (Grad Level)

- Assisted with class preparation and providing feedback

The University of Electro-Communications

Tokyo, Japan

First-year Students Learning Assistant

Apr. 2021 – Mar. 2022

- Assisted first-year students in acclimating to college life and classes during the COVID-19 pandemic

SKILLS & LANGUAGES

- Technical: Python, C, Processing, Git, Unity, Arduino/ESP32, KiCad, Fusion 360, 3D Printing.
- Languages: Korean (Native), Japanese (Fluent), English (Advanced)